

# ThermoBlinds: Non-Contact, Highly Responsive Thermal Feedback for Thermal Interaction

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Figure 1: System overview (left: the shutter opens to let infrared rays to pass through for thermal feedback, middle: the shutter closes to block infrared rays, so there is no thermal feedback) and a superimposed visual and thermal media experience (right).

## ABSTRACT

We present *ThermoBlinds*, a non-contact, highly responsive thermal feedback device for thermal interactions. It provides responsive feedback by rapidly adjusting infrared irradiance with a shutter mechanism. The feedback can be applied according to visual media contents to reproduce thermal experiences. By providing thermal feedback in accordance with the user's gaze, it is possible to create a visual media experience with a high resolution thermal sensation. Additionally, it supports remote communication by providing thermal feedback based on the other person's gaze.

## KEYWORDS

infrared rays, warm sensation, thermal feedback

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## 1 INTRODUCTION

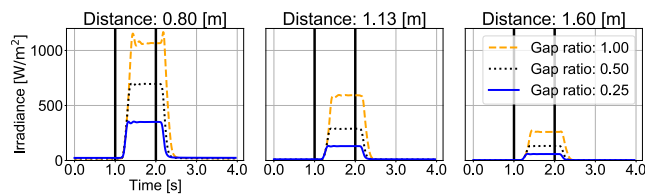
Thermal sensation is an important aspect of human perceptual and emotional experience. We perceive our environment without contact through thermal sensations as well as audiovisual sensations. For example, we sense the sun from the warmth of sunlight and recognize seasons from ambient temperature. Thermal sensation can also induce feelings of comfort or discomfort. The use of thermal feedback in human-computer interaction (HCI) has the potential to become a new mode of enhancing realism and presence or arousing emotions. As in the physical world, presenting thermal sensations is expected to enrich the audiovisual perceptual experience or create a unique experience on its own. Application of non-contact thermal feedback in HCI has not been well studied. One reason for this is that conventional non-contact thermal feedback methods are less responsive than contact thermal feedback methods, making them unsuitable for many interactive applications.

In this paper, we propose *ThermoBlinds* (Figure 1), a non-contact, highly-responsive thermal feedback device, with an infrared source and a shutter mechanism that controls infrared irradiance quickly. Then, we demonstrate its applications, including a media experience where the thermal feedback corresponds with the visual contents, vision augmentation where the user feels the temperature of the object they see, and remote communication support where the user feels the other user's gaze as heat.

## 2 DESIGN AND IMPLEMENTATION

*ThermoBlinds* consists of an infrared source and a shutter mechanism with four rotating polycarbonate fins (Fig.1). The infrared

source (CORONA DH-1220R) emits infrared rays with wavelengths ranging from 3 to 20  $\mu\text{m}$  without causing air flow. The shutter mechanism controls the infrared irradiance continuously over a wide range. The approach is similar to that of Wareing et al. [Wareing et al. 2018], but we use fins to change the irradiance continuously without significantly changing the area where thermal feedback is presented. Aluminum tape was placed on one side of each fin to ensure that the infrared rays did not transmit through the fins. Each fin was rotated using a servo motor (Futaba RS204MD), controlled by a microcontroller (Arduino Uno R3). The width, height and depth of the device were 500 mm, 780 mm, and 700 mm, respectively. The total wattage required is 1.15 kW. The system controls thermal perception within about 0.5 s regardless of the distances to the users (Figure 2).



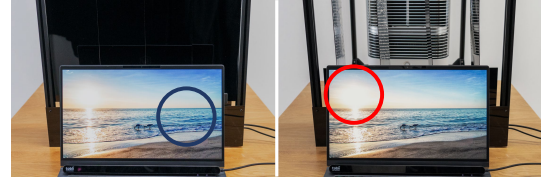
**Figure 2: Measured infrared irradiance at different distances. The thick vertical lines show the time when controlling signals were sent to open and close the shutter mechanism.**

### 3 APPLICATIONS

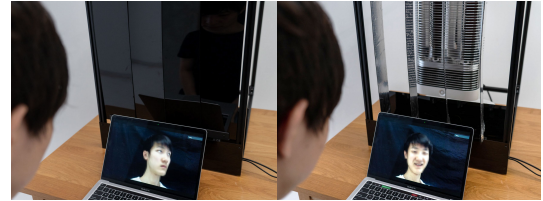
ThermoBlinds can be used as a device to provide responsive thermal feedback to augment visual media and to implicitly change people's behavior by controlling their thermal comfort. In addition, it supports remote communication by emphasizing gaze of the other user with thermal feedback. Here, we introduce three applications using ThermoBlinds.

First, we have developed a visual media experience with thermal feedback. Conventional visual media experiences are not accompanied by thermal sensations, and it reduces the realism of the contents. For example, when we see a flame in the physical world, we feel the heat, but we cannot feel the heat from the flame presented on the monitor. To solve this problem, we projected the visual feedback on an infrared transmissive mesh screen and put ThermoBlinds behind the screen so that the visual and thermal feedback are superimposed (Figure 1 Right). In addition, by providing thermal feedback based on the user's gaze, the user perceives heat from individual objects they are looking at in visual media. For example, they perceive a hot sensation when they look at the sun, and then when they look at the water, they perceive cooling sensation owing to the loss of heat given to them. (Figure 3).

Next, we have developed a remote communication support system using gaze-based thermal feedback. In conventional remote communication, gaze of the other person is recognized only by limited visual information through the monitor. We developed a system to support remote communication using thermal feedback and eye tracking to express other's gaze with thermal feedback as shown in Figure 4. A user feels warmth when the other user is looking directly at them, and when the direct gaze is averted from



**Figure 3: Gaze-based Thermal Feedback for Visual Media.**



**Figure 4: Remote communication support system using temperature-based gaze presentation.**

them, the user feels a relative coolness due to the loss of warmth. The system allows users to communicate with each other remotely while feeling the presence of the other person as if they were face-to-face. Moreover, they can even perceive the other's gaze when they are not looking at the monitor.

Finally, non-contact thermal feedback can implicitly change a person's behavior. People sometimes behave in ways that are different from what is good for them or what they want to do. For example, they often find it difficult to concentrate on their work due to distractions, or have poor posture after sitting for a long time. Subtle activity-specific thermal feedback can help solve these problems by implicitly influencing people's behavior. As Narumi et al. [Narumi et al. 2009] have shown, people tend to change their behavior to become thermally comfortable. Therefore, non-contact, subtle thermal feedback has the potential to change people's behavior without interfering with their activities by adjusting their comfort level. For example, if a person has a good posture, they will get comfortable thermal feedback, but if their posture gets worse, they will lose thermal feedback. In this way, the person implicitly keeps a good posture to be thermally comfortable.

At the event, attendees will experience (A) superimposed visual and thermal media (Figure 1 right) and (B) gaze-based thermal feedback for visual media (Figure 3). In (A), attendees will watch a video accompanied by thermal feedback. In (B), attendees take a chair in front of a monitor and ThermoBlinds, and then, explore the visual content on the monitor with gaze-based thermal feedback.

### ACKNOWLEDGMENTS

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